Time Synchronization of OpenSky's ADS-B Receivers

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Outline

1. Motivation
2. Approach
3. Data Analysis
4. Results

Motivation

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  - "The OpenSky Network is a community-based receiver network which continuously collects air traffic surveillance data"

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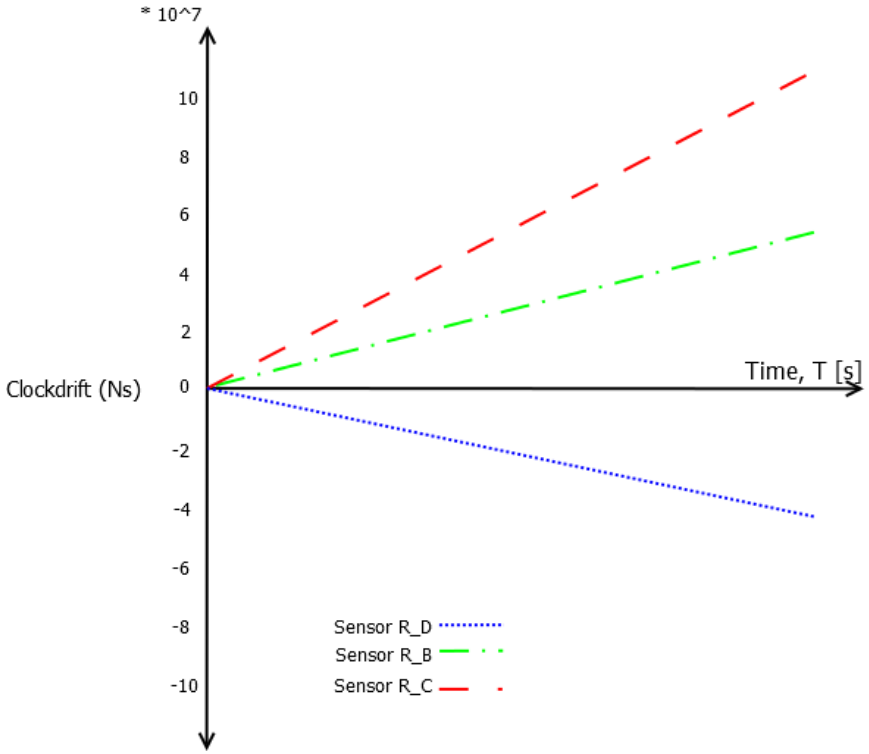
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  - "The OpenSky Network is a community-based receiver network which continuously collects air traffic surveillance data"
  - Multilateration requires three or more synchronized sensor nodes

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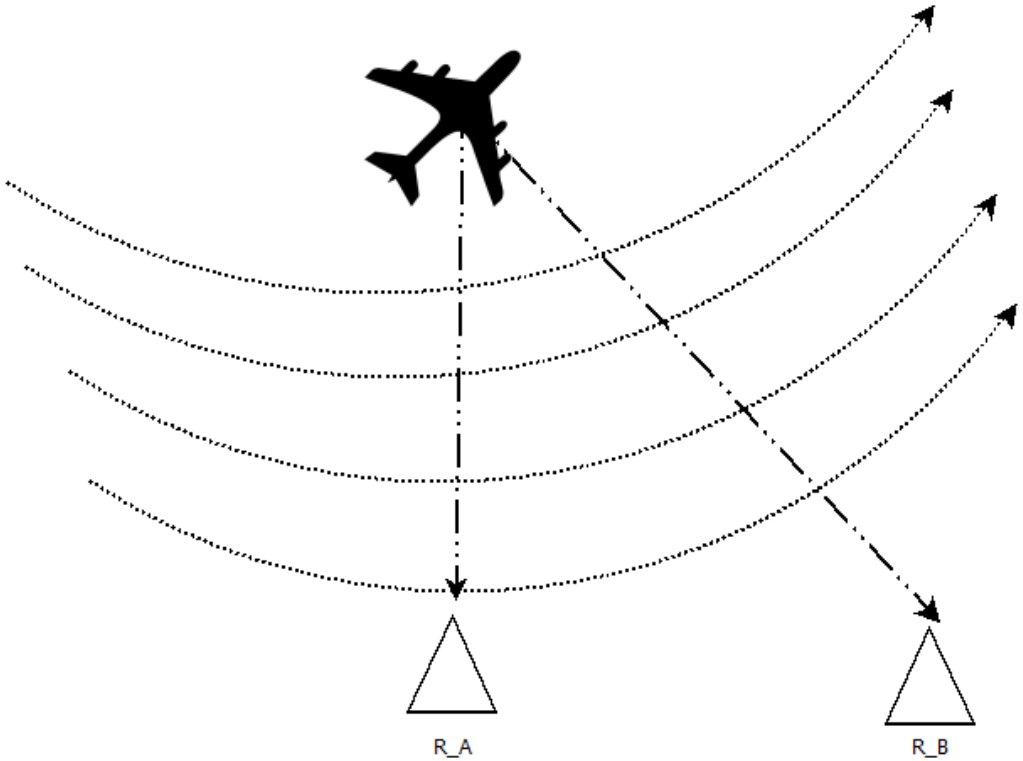
- Why we need time synchronization in OpenSky?
  - "The OpenSky Network is a community-based receiver network which continuously collects air traffic surveillance data"
  - Multilateration requires three or more synchronized sensor nodes
  - achieve synchronization within the range of nanoseconds

Motivation

- To synchronize the network:
  - Find the synchronization error and synchronize the node or,
  - Predict the offset



Approach



- Let R_A and R_B be two receivers and let i be a location claim event sent by the flight,
- t_A^i and t_B^i denotes the reception timestamp of the location claim event i.

Approach

- Δ_i^A and Δ_i^B denotes the propagation delay between receiver R_A and aircraft, receiver R_B and aircraft respectively

Ground Truth

$$\begin{aligned} \Phi_i^{AB} = \Omega_i^{AB} - \theta_i^{AB} &= (t_i^A - t_i^B) - (\Delta_i^A - \Delta_i^B) \\ &= (t_i^A - \Delta_i^A) - (t_i^B - \Delta_i^B) \end{aligned} \tag{1}$$

- where $\Omega_i^{AB} = t_i^A - t_i^B$,
- $\theta_i^{AB} = \Delta_i^A - \Delta_i^B$

Hardware details

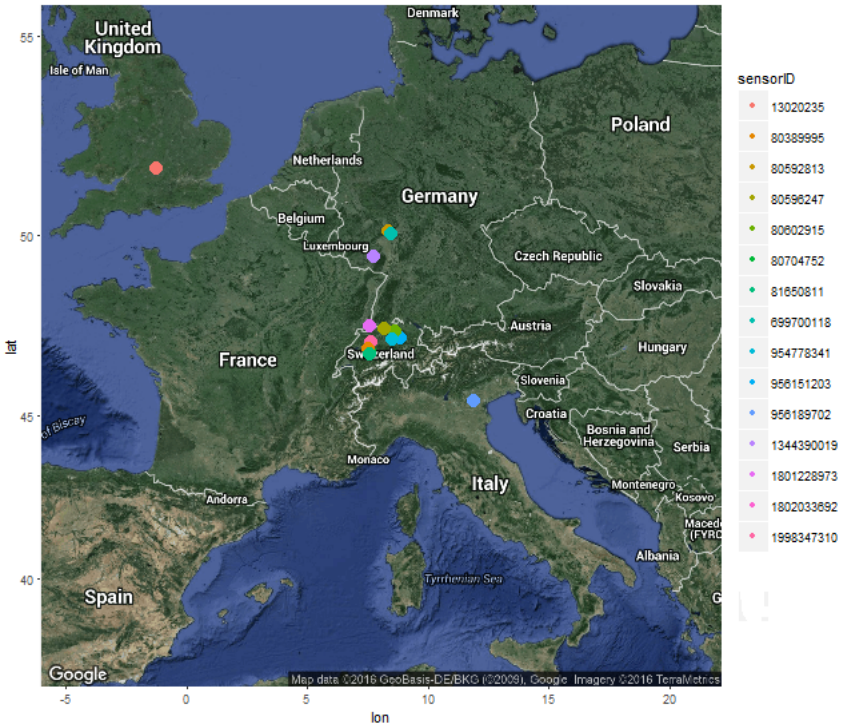
- Radarcape receivers
- 30 bit rolling timestamps nanosecond precision.

Preprocessing

- Analyzed two different data sets, from different point in time
- Data set includes message sensors which are GPS synchronized and not

Preprocessing

- Filtered the data from multiple reception
- Choose the duplicate position reporting messages
- Duplicates are position messages sent from an aircraft that are received by two or more sensors



How to solve it??

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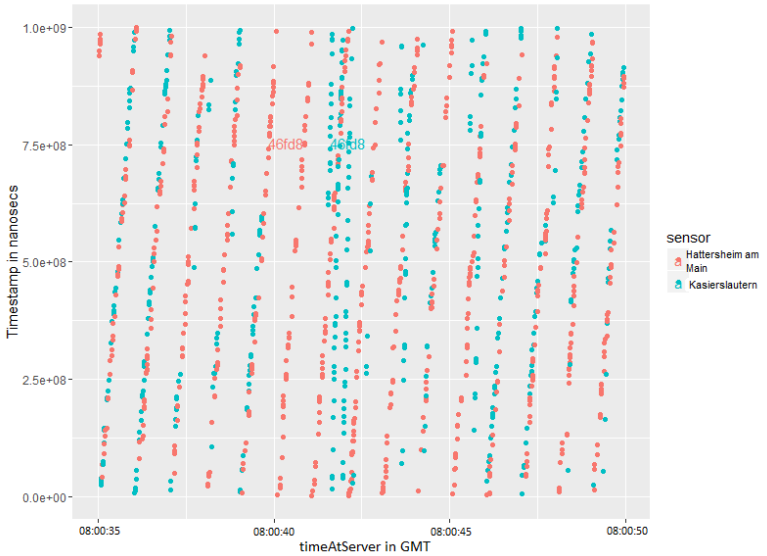
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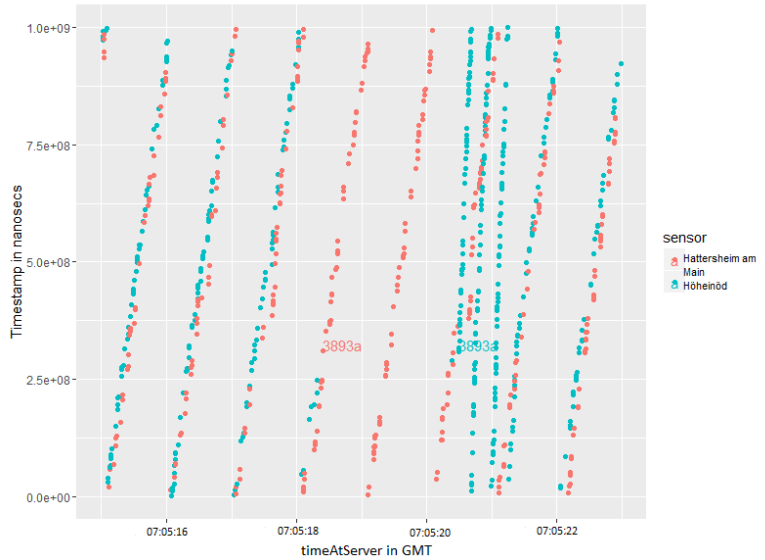
How to solve it??

- Internal synchronization
- Assume one sensor node as the master node and synchronize the clock of other nodes with the master node or,
- predict the offset of sensor node for the time t with to the master node
- and why not other methods?

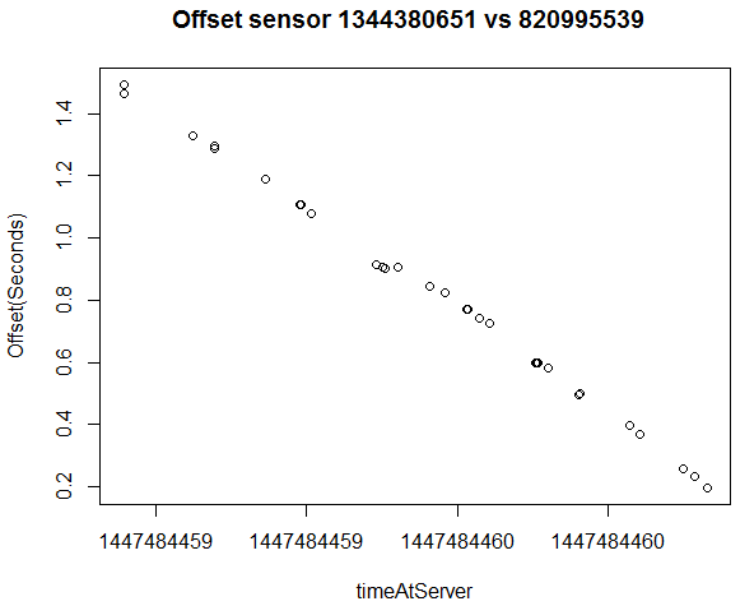
Results : Unexpected IJ for duplicates

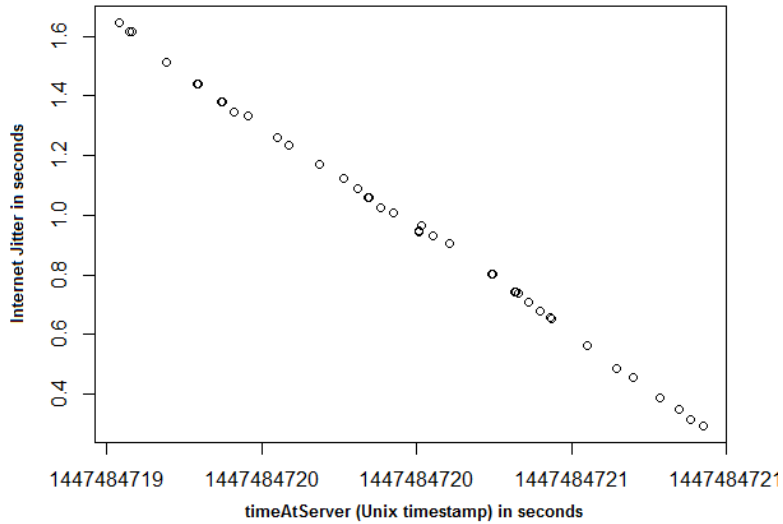
- Unexpected IJ for duplicates.





Results : Unexpected IJ for duplicates





- Typical TCP phenomenon

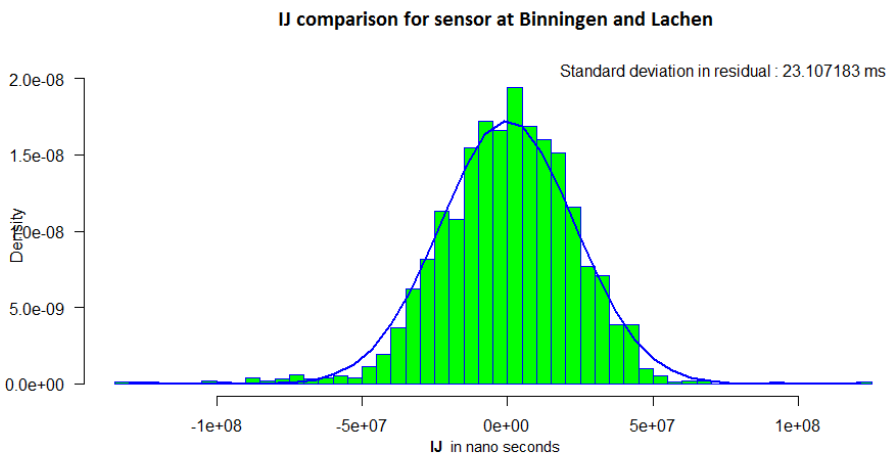
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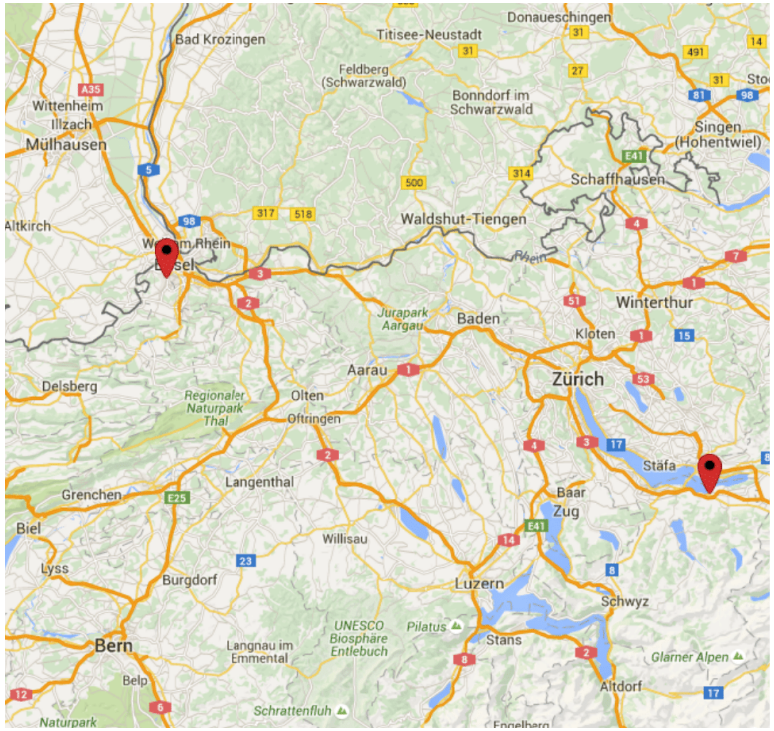
- *timeAtSensor* was supposed to provide the second of each day

Results

- *timeAtSensor* was supposed to provide the second of each day
- But, false readings for some sensors!!!

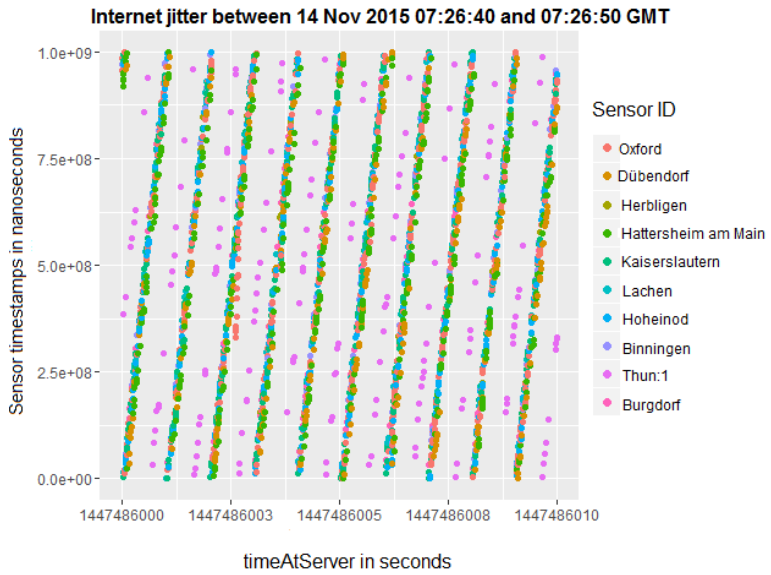
Results


IJ comparison for sensor at Binningen and Lachen
Standard deviation in residual : 23.107183 ms
2.0e-08
1.5e-08
1.0e-08
5.0e-09
0.0e+00
Density
-1e+08
-5e+07
0e+00
5e+07
1e+08
IJ in nano seconds


Donaueschingen
Bad Krozingen
Titisee-Neustadt
Feldberg (Schwarzwald)
Bonndorf im Schwarzwald
Wittenheim
Illzach
Mülhausen
Singen (Hohentwiel)
Schaffhausen
Waldshut-Tiengen
Altkirch
Rhein
Winterthur
Jurapark Aargau
Baden
Kloten
Aarau
Zürich
Delsberg
Regionaler Naturpark Thal
Olten
Oftringen
Stäfa
Langenthal
Grenchen
Baar
Zug
Biel
Willisau
Lyss
Burgdorf
Luzern
Schwyz
UNESCO Biosphäre Entlebuch
Pilatus
Bern
Langnau im Emmental
Stans
Glarner Alpen
Belp
Altdorf
Schrattenfluh
Naturpark

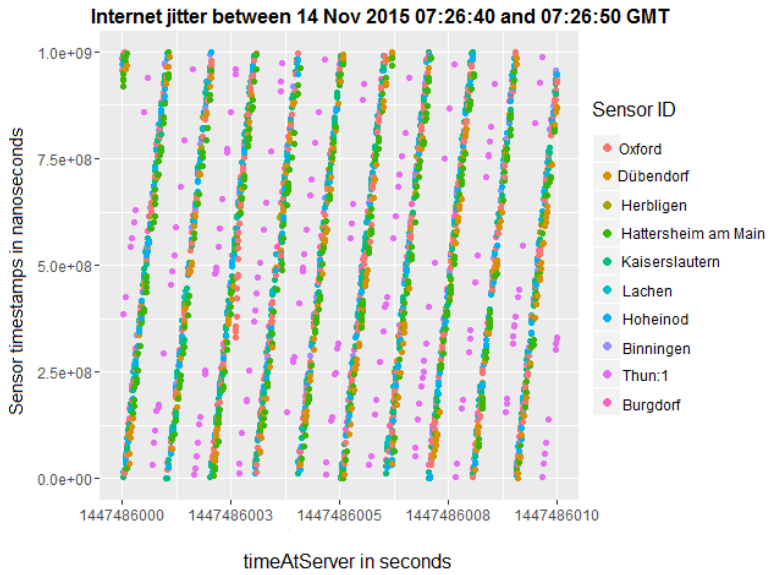
Results

- checked the Internet Jitter (IJ) for every message received by the sensor pair to reach the server.



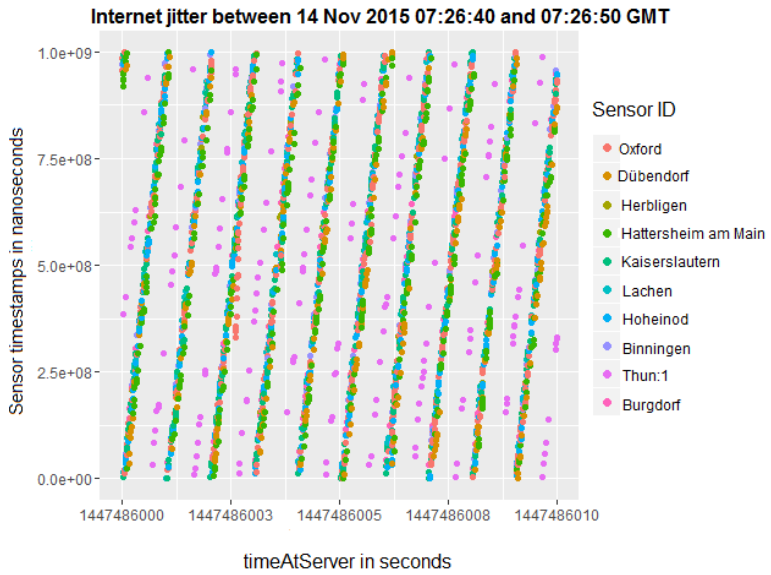
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- IJ was in the range of microseconds.



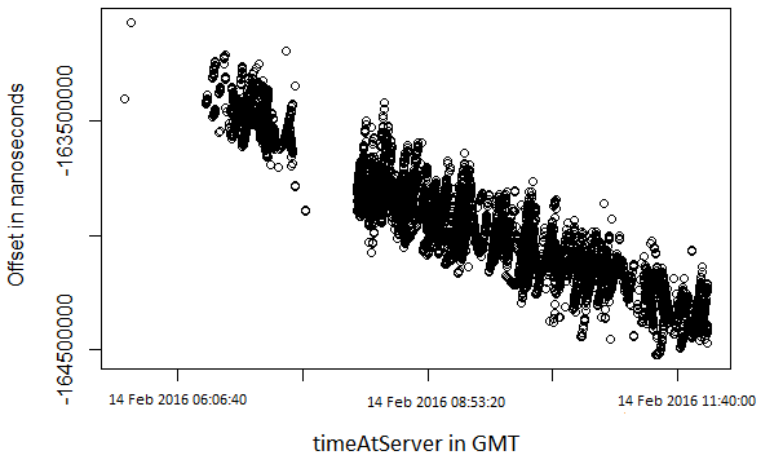
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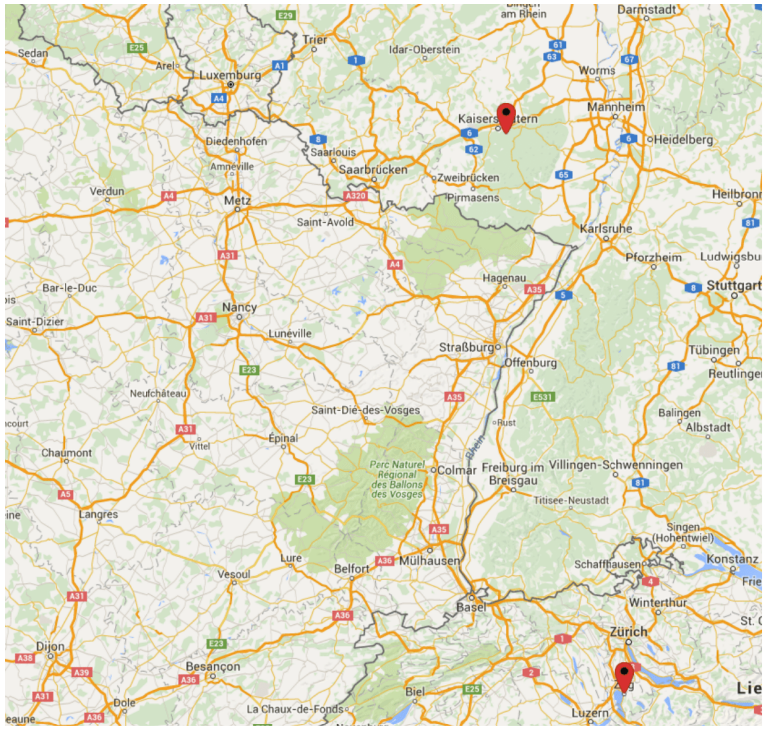
- checked the Internet Jitter (IJ) for every message received by the sensor pair to reach the server.
- IJ was in the range of microseconds.
- So, compare local timestamp counter of each sensor



Results : Sensor without GPS sync

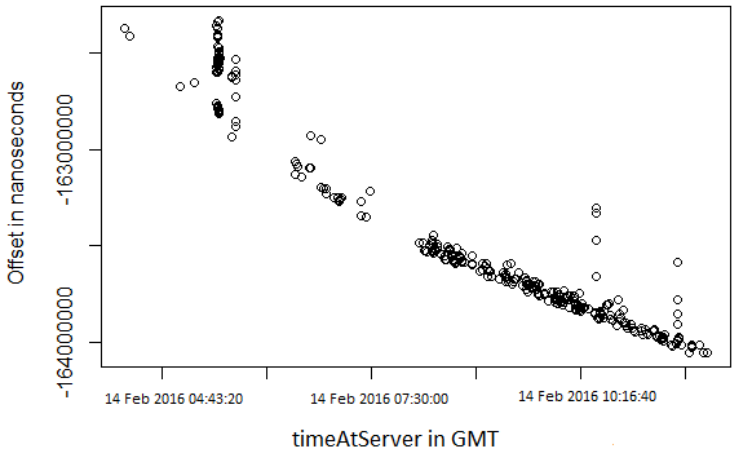
- Sensor pair Zug and Uni-KL

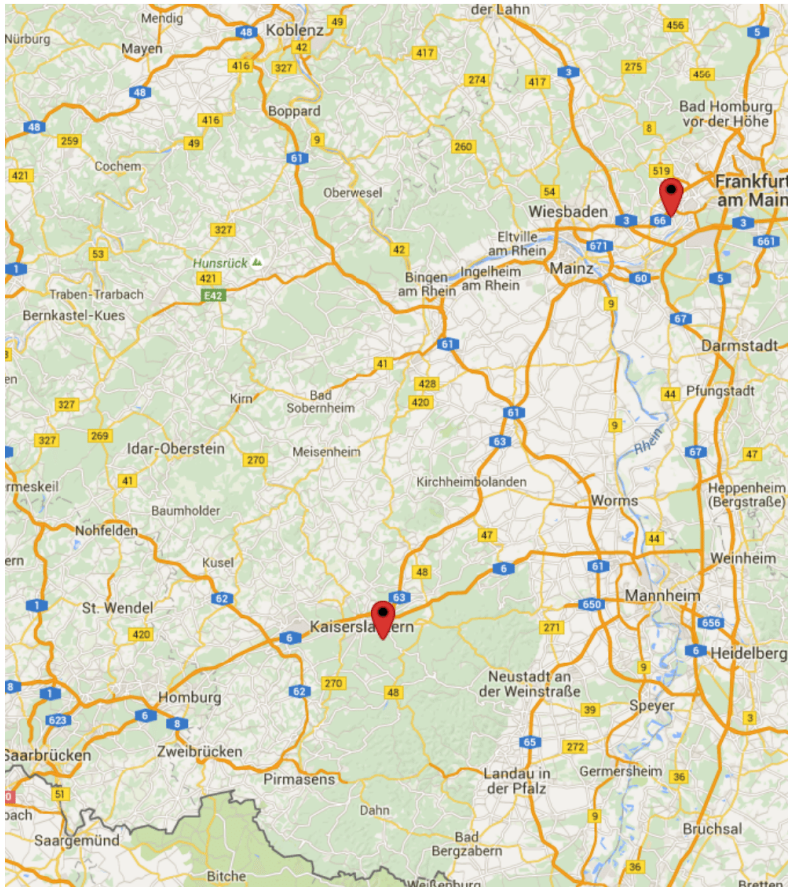




Results : Sensor without GPS sync

- Sensor pair Hattersheim am Main and Uni-KL

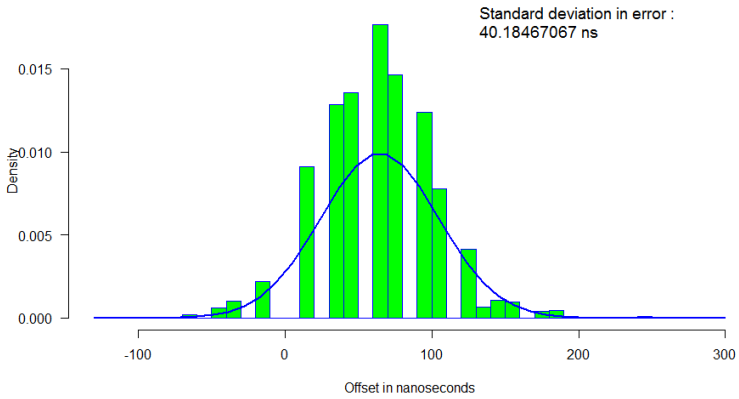


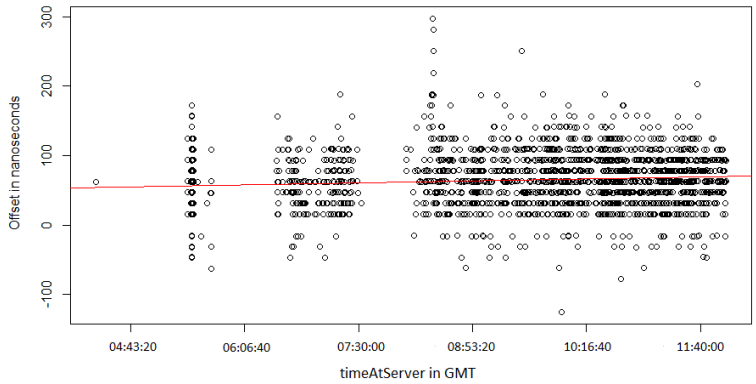


- Shows that sensor at Uni-KI was not GPS synchronized

Results

- Sensor at Thun

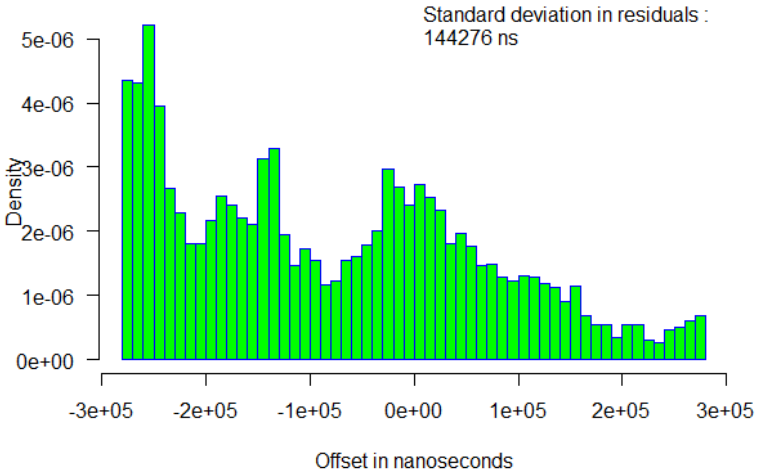


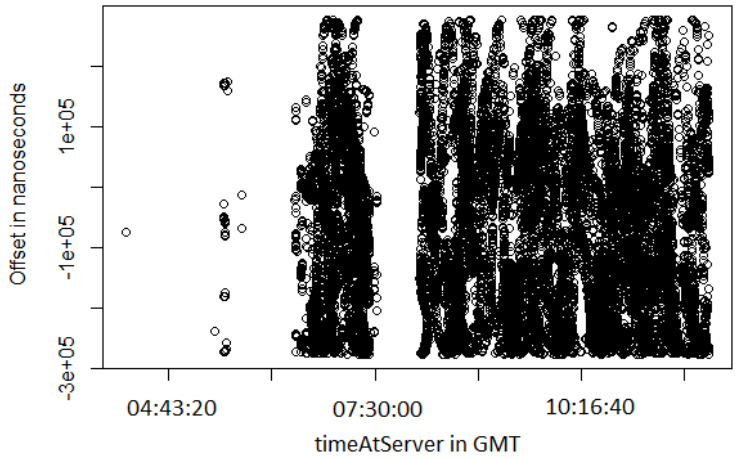


- GPS synchronized, also shows the precision of Radarcape timestamp counter (15 to 16 ns)

Results

- Sensors at Zug and Binningen (100 km apart)





Results

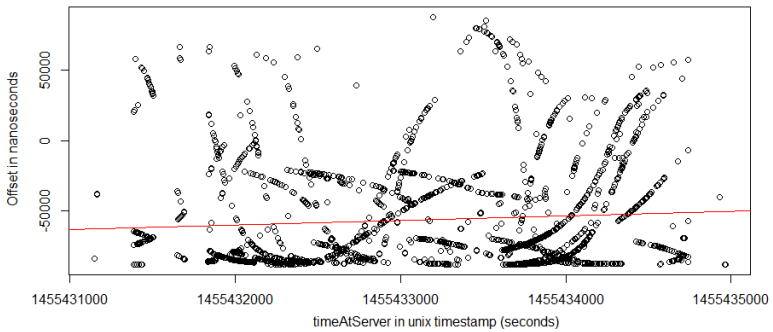
- Repeated the process for other sensor pairs

Results

- Repeated the process for other sensor pairs
  - Minimum offset in synchronization error with standard deviation of 40.185 ns (0.016 km apart, sensors at Thun)
  - Maximum offset in synchronization error with standard deviation of 144276 ns (192.16 km apart, Zug and Binningen)

Results

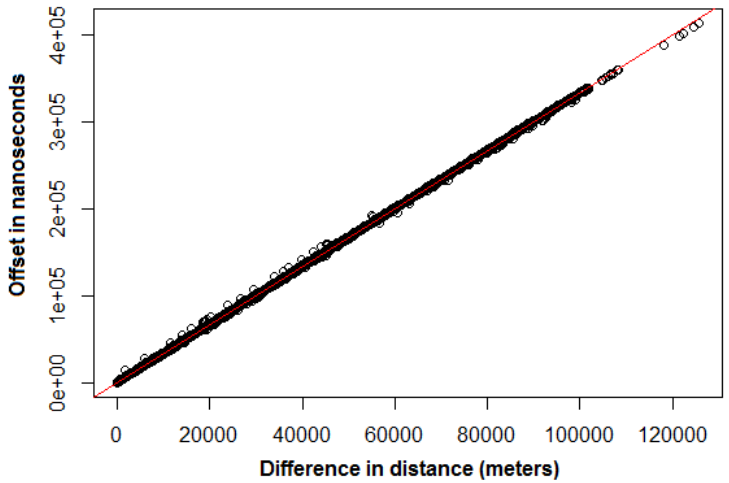
- Plot showing offset at a certain timestamp for sensor at Dübendorf with respect to the sensor at Zug



- shows that there is error in Propagation delay calculation, i.e. error is distance calculation

Results

- World Geodetic System 1984 (WGS 84), the GPS model of earth
- To calculate the error, consider two GPS synchronized nodes and measure the offset with respect to increase in distance.
- Propagation delay with respect to difference in distance by comparing sensors at Hattersheim am Main and Zug



Results

- Modified equation for an event i for sensors A and B

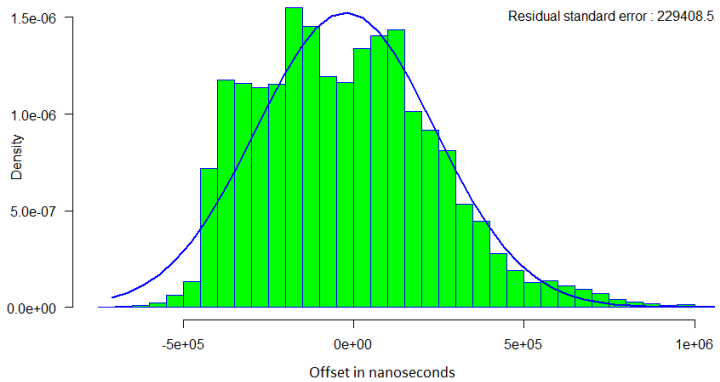
Final equation

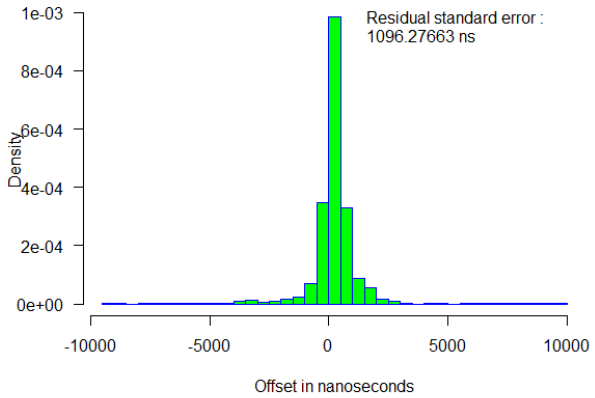
$$\Phi_i^{AB} = \Omega_i^{AB} - \theta_i^{AB} = (t_i^A - t_i^B) - (\Delta_i^A - \Delta_i^B) - Error_d \quad (2)$$

- Δ_i^A is the propagation delay for the message to reach from aircraft to R_A
- Δ_i^B is the propagation delay for the message to reach from aircraft to R_B

Results

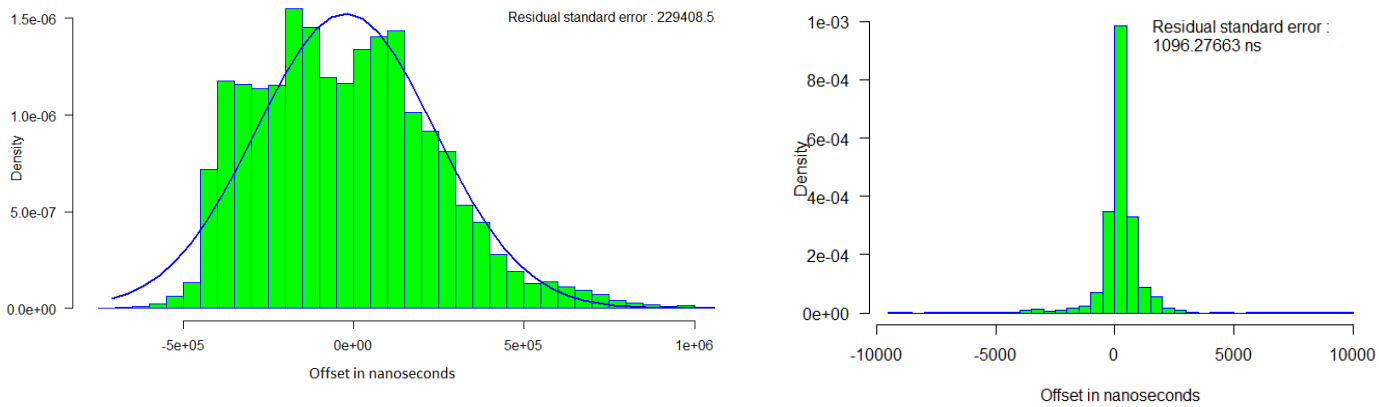
- Improvement in standard deviation for synchronization error
- Histogram showing residual distribution Hattersheim am Main (Frankfurt) vs Zug (229408.5283ns to 1096.2766ns)





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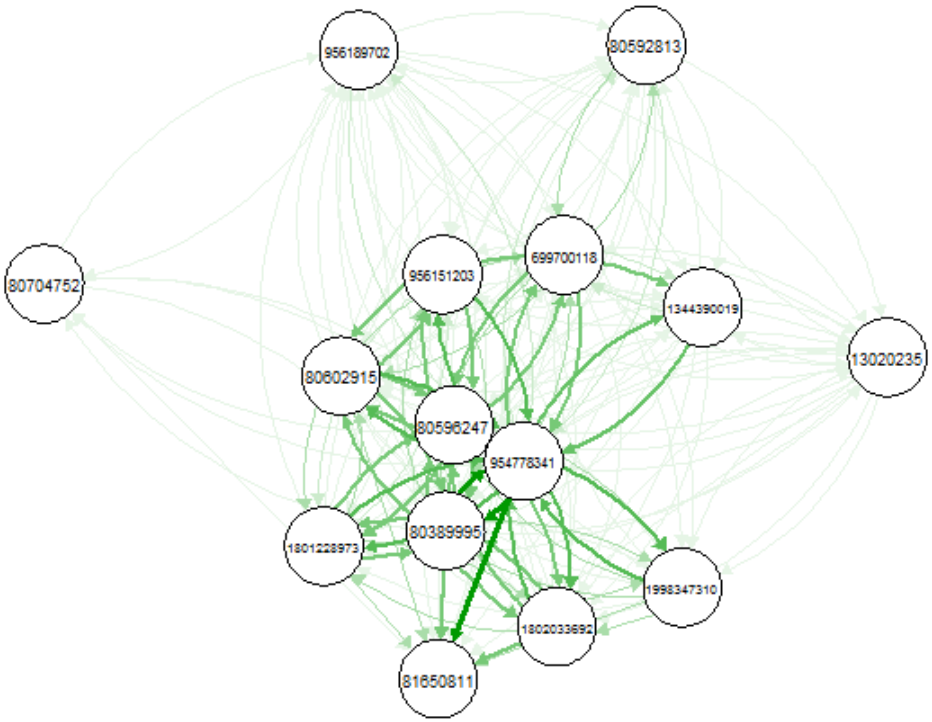


- But, 1096.2766ns means an error of 329 meters. Is it a good result for air-traffic surveillance??

Results

- Lets synchronize!
- consider the *tuples* or *pairs*

Result : How we sync ?



- Sensor node *954778341* placed at Zug was considered as master.
- Measured the offset the for each pair

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- Still error exits, and the source for those are ;
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  - pressure altitudes vs true altitude
  - inaccuracy in the position information of the receivers

Summary

- Goal : analyze the time synchronization of the sensor nodes in the OpenSky network.
- Findings,

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- Findings,
  - tried out different ways to achieve synchronization,
  - fault timeAtSensor,
  - unsynchronized nodes in the network,
  - how could we improve our result.


Thank You